RevoU FSDA Batch FEB26

TokoBli Dataset

TokoBli Campaign Evaluation & Product Page Analysis

Optimizing Campaign Strategy and Product Experience

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Business Background

Business Background

TokoBli is one of the largest e-commerce platforms in Indonesia that regularly runs promotional events during **twin date campaigns** such as **10.10, 11.11, and 12.12** to increase sales and attract more customers.

However, these campaigns require significant **discount budgets** and promotional costs. Therefore, it is important to evaluate **past campaign performance** to understand which campaign strategy delivers the best results and how **future campaigns can be optimized**.

General Performance

During the observed period, the platform generated **Rp46.9B in total revenue from 10.153 transactions** , indicating strong overall sales activity.

Customer purchasing behavior shows that most transactions involve **only one product per order** , with an average quantity of **1.28 items per transaction** .

In terms of promotions, a large portion of purchases occur **without discounts** , as reflected by the **Rp0 median discount value**. This suggests that many customers are willing to buy products even without promotional incentives.

Meanwhile, the typical transaction value is around **Rp4.27M** , while a small number of higher-value transactions contribute significantly to overall revenue.

Total Revenue :

Rp46,908,982,000

Total Transaction:

10.153

Average Quantity per Transaction :

1.28 items

Average Discount per Transaction :

Rp82,097

Problem Statement – SMART

Which twin date campaign period generates the best business performance based on transaction quantity, discount spending, and total revenue?

Metrics to evaluate :

- Total Revenue
- Quantity per Transactions
- Discount Value

02

Data Preparation

Change Summary

Column(s)	Changes	Note(s)
Status	Removed column	More than 50% missing values
SKU	Removed column	Not relevant for analysis
Shipping Cost	Removed column	All values were 0
Price, Discount, Total Revenue	Formatted to currency	Standardized monetary format
Created At	Standardized date format	Converted to readable date format
Price, QTY, Discount	Missing values imputed	Calculated using transaction relationships
All	Removed duplicate rows	5 duplicate records detected and removed
All	Removed empty rows	5 rows contained no data
Total Revenue	Removed outlier	Impacted : ID-451

Descriptive Statistics – QTY

QTY

Descriptive Statistics	Value
Mean	1.28
Standard Error	0.01
Median	1
Mode	1
Standard Deviation	1.11
Sample Variance	1.24
Kurtosis	370.66
Skewness	16.10
Range	30
Minimum	1
Maximum	31
Sum	12965
Count	10153
Q1	1
Q3	1

- **Most customers buy only one item per transaction**
  The **average** quantity per transaction is **1.28**, while both the **median and mode** are **1**. This shows that most customers typically purchase **just one product in each transaction**.

- **The distribution is highly right-skewed**
  The skewness value (**16.10**) indicates that most transactions involve small quantities, while only a few transactions contain larger quantities. The maximum quantity reaches **31 items**, but these cases are very rare.

- **Most transactions are concentrated at the lowest quantity**
  The quartile values (**Q1 = 1 and Q3 = 1**) show that the majority of transactions involve only **one item**, meaning customers rarely purchase multiple items in a single order.

Conclusion :
Overall, the data suggests that **single-item purchases dominate customer behavior** on the platform. Only a small portion of transactions involve multiple items.

Descriptive Statistics – Discount

Discount

Descriptive Statistics	Value	
Mean	Rp82,097	
Standard Error	Rp1,859	
Median	Rp0	
Mode	Rp0	
Standard Deviation	Rp187,279	
Sample Variance	Rp35,073,423,155	
Kurtosis	Rp5	-Rp4,472
Skewness	Rp3	
Range	Rp900,000	
Minimum	Rp0	
Maximum	Rp900,000	
Sum	Rp833,527,000	
Count	10,153	
Q1	Rp0	
Q3	Rp21,000	

- **Most transactions do not use discounts**
  The **median and mode** are both **Rp0**, which indicates that the majority of transactions happen **without any discount applied** .

- **The distribution is highly right-skewed**
  The skewness value (**3**) suggests that most discounts are relatively small, while only a few transactions receive **very large discounts** .

- **Large variation in discount values**
  The discount ranges from **Rp0 to Rp900,000** , with a standard deviation of **Rp187,279** , showing that the size of discounts varies widely across transactions.

Conclusion :
Overall, the data shows that most purchases are made without discounts, while only a small number of transactions receive larger promotional incentives.

Descriptive Statistics – Revenue

Total Revenue

Descriptive Statistics	Value
Mean	Rp4,620,209
Standard Error	Rp32,318
Median	Rp4,275,000
Mode	Rp7,200,000
Standard Deviation	Rp3,256,399
Sample Variance	Rp10,604,135,976,592
Kurtosis	Rp127
Skewness	Rp7
Range	Rp78,355,000
Minimum	Rp45,000
Maximum	Rp78,400,000
Sum	Rp46,908,982,000
Count	10,153
Q1	Rp2,241,000
Q3	Rp6,980,000

- **Most customers spend around Rp4.27M per transaction**
  The median transaction value is **Rp4,275,000** , which shows the typical amount customers spend in one purchase. The mean (**Rp4,620,209**) is slightly higher, meaning a few larger transactions push the average up.

- **A few transactions generate much higher revenue**
  The skewness value (7) indicates that while most transactions fall within the lower to mid revenue range, a small number of purchases reach much higher values.

- **Customer spending varies quite widely**
  Transaction values range from **Rp45,000 to Rp78,400,000** , showing a wide gap between smaller and larger purchases.

Conclusion :
Overall, most customers make **moderate-sized purchases** , while a small number of high-value transactions contribute a large share of the total revenue.

Key Insight

- **Customers usually buy only one item per order**
  Most transactions involve a single product, showing that customers typically come to purchase a specific item rather than multiple products.

- **Many purchases happen without discounts**
  More than half of the transactions do not use any discount, meaning customers are still willing to buy products at the regular price.

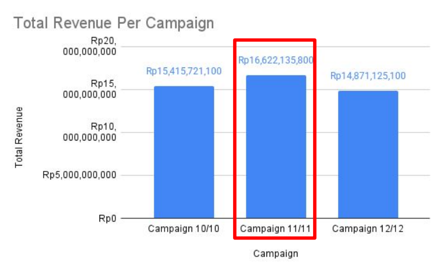
- **A small number of purchases generate very high revenue**
  While most transactions fall within a moderate spending range, a few high-value purchases contribute significantly to the total revenue.

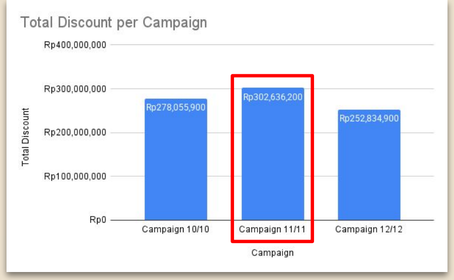
03

Statistical Analysis

Campaign Performance Comparison

Campaign	Total Transaction	Total Customer	Total Product Sold	Total Revenue	Total Discount
Campaign 10/10	3339	1504	4355	Rp15,415,721,100	Rp278,055,900
Campaign 11/11	3399	1567	4115	Rp16,622,135,800	Rp302,636,200
Campaign 12/12	3415	1448	4495	Rp14,871,125,100	Rp252,834,900





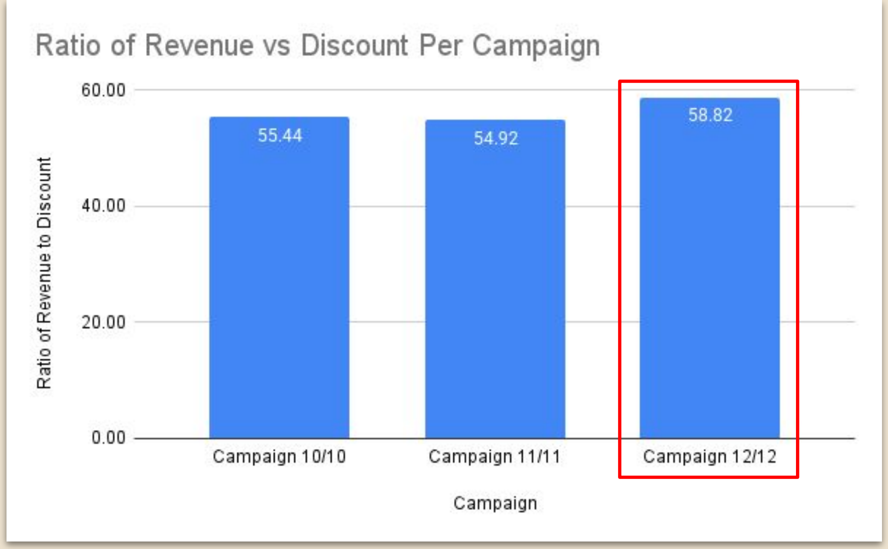
Campaign 11.11 has the **highest revenue**, but it also comes with the **highest discount** spending, indicating that its **strong performance is largely driven by aggressive promotions**.

Meanwhile, **Campaign 12.12** records the **highest number of transactions and products sold**, but generates **lower revenue** than Campaign 11.11. This suggests that **higher sales volume does not always translate into higher revenue**.

Going forward, it is important to **optimize discount strategies** to **maintain strong revenue** while **improving cost efficiency**.

Campaign Efficiency Analysis

Campaign	Total Revenue	Total Discount	Ratio
Campaign 10/10	Rp15,415,721,100	Rp278,055,900	55.44
Campaign 11/11	Rp16,622,135,800	Rp302,636,200	54.92
Campaign 12/12	Rp14,871,125,100	Rp252,834,900	58.82



From an efficiency perspective, **Campaign 12.12** stands out as it generates the **highest return relative to its discount spending**. This indicates a more **optimal use of promotional budget** compared to the other campaigns.

On the other hand, **Campaign 11.11** appears **less efficient**, as a larger portion of its revenue is driven by **higher discount allocation**.

This suggests that future **campaigns** should **focus** on smarter **discount allocation** rather than increasing promotional spending, in order **to achieve better returns** while maintaining strong performance.

Product Category Performance

Men's Fashion stands out as the **top-performing category**, contributing the **highest revenue and transactions**.

Meanwhile, categories like **Superstore** and **Health & Sports** also show strong performance with **lower discount usage**, indicating **better efficiency**.

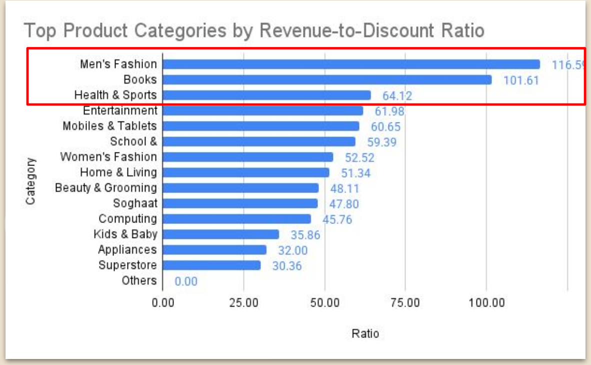
Moving forward, **Men's Fashion** can continue to be leveraged as the **main revenue** driver with a more optimized discount strategy, while greater **focus** can be given to **Superstore** and **Health & Sports** to support more **efficient growth**.

Campaign 11/11	
Men's Fashion	Rp8,246,126,000
Superstore	Rp2,040,230,000
Health & Sports	Rp1,074,092,000
Beauty & Grooming	Rp1,010,663,800
Home & Living	Rp980,455,000
Women's Fashion	Rp902,309,000
Appliances	Rp701,922,000
Mobiles & Tablets	Rp637,921,000
Kids & Baby	Rp388,691,000
Soghaat	Rp362,828,000
Computing	Rp157,050,000
Entertainment	Rp72,540,000
School & Education	Rp33,202,000
Books	Rp14,106,000

Product Category Efficiency

Category All Campaign

Category Name	Total Revenue	Total Discount	Ratio
Men's Fashion	Rp14,693,964,000	Rp126,036,000	116.59
Books	Rp105,977,000	Rp1,043,000	101.61
Health & Sports	Rp4,207,256,000	Rp65,614,000	64.12
Entertainment	Rp341,599,000	Rp5,511,000	61.98
Mobiles & Tablets	Rp2,720,082,000	Rp44,848,000	60.65
School & Education	Rp301,666,000	Rp5,079,000	59.39
Women's Fashion	Rp4,127,341,000	Rp78,579,000	52.52
Home & Living	Rp3,471,258,000	Rp67,619,000	51.34
Beauty & Grooming	Rp6,094,399,800	Rp126,671,200	48.11
Soghaat	Rp2,108,225,000	Rp44,109,000	47.80
Computing	Rp782,882,000	Rp17,108,000	45.76
Kids & Baby	Rp1,369,652,000	Rp38,198,000	35.86
Appliances	Rp1,976,429,000	Rp61,761,000	32.00
Superstore	Rp4,595,751,200	Rp151,350,800	30.36
Others	Rp12,500,000	Rp0	0



Men's Fashion stands out as the **most efficient category**, generating the **highest return from its discount spending**. Other categories like **Books and Health & Sports** also show **strong efficiency**, indicating that they can drive solid performance **without relying heavily on discounts**.

This suggests that some categories naturally perform better even with lower promotional support. Going forward, it would be more effective to **prioritize these high-efficiency categories** and **allocate discounts more strategically**, so the business can maximize results without increasing costs unnecessarily.

A/B Testing Analysis: Product Page Comparison

Hypothesis Question

Is there a significant difference in transaction value between the current product page and the new product page?

Hypothesis

H0 : TVal control (old page) = TVal treatment (new page)
H1 : TVal control (old page) <> TVal treatment (new page)

Test & Assumptions

Test : Two-sample T-Test (Equal Variances)
Alpha : 5%
Assumptions : Variance ratio ≤ 4 → equal variance assumpted

Result

Control Mean : 746,102.6140
Treatment Mean : 830,460.1594
P-Value : < 0.05 → Rejected H0
TVal Control < TVal Treatment

	Control	*Treatment*
Mean	746102.6104	830460.1594
Variance	22436486774	33088798170
Observations	498	502
Pooled Variance	27783989789	
Hypothesized Mean Difference	0	
df	998	
t Stat	-8.001901057	
P(T<=t) one-tail	0	
t Critical one-tail	1.646381816	
P(T<=t) two-tail	0	
t Critical two-tail	1.962343802	

A/B Testing Insight & Business Recommendation

Observation
The new product page shows a **higher average transaction value** compared to the current design.

Business Impact
This indicates a strong **potential to increase** overall **revenue** through improved user experience.

Insight
The difference is **statistically significant** , meaning the improvement is not do to chance.

Prioritization
High – directly impacts revenue and user spending behavior.

Recommendation
Roll out the new product page to all user and continue monitoring its performance after implementation.

04

Campaign Performance Insight

Final Insight & Business Recommendation

Campaign 11.11 delivers the **highest revenue and transactions**, but it also comes with the **highest discount spending**. On the other hand, **Campaign 12.12** shows **better efficiency**, as it can generate solid results with a lower budget. This shows that strong performance is not just about spending more, but about **using the budget more effectively**.

Looking at the categories, products like **Men's Fashion and Health & Sports** consistently perform well and generate **higher returns without relying too much on discounts**. This suggests that certain categories already have strong demand and can be further leveraged.

On top of that, the A/B testing results show that the **new product page increases transaction value**, meaning that a **better user experience** can directly drive higher spending.

Going forward, it would be more effective to focus on **optimizing discount allocation, prioritizing high-performing categories,** and **rolling out the new product page** to support more sustainable growth.

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THANK YOU!

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